STEAMROOT · RISK REGISTER

SteamRoot risk register and hazard analysis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade): project risks, preliminary hazard analysis and first-pass design FMEA, from the safety sections of STR-PRC-001 and STR-BLD-001, STR-STM-001 and STR-CAL-001 v0.5

SteamRoot risk register and hazard analysis

The top project risk is regulatory: if the Texas boiler authority does not accept the open-vented design under the atmospheric-pressure exemption, SteamRoot cannot be built as conceived (R1). The top hazards are steam burns at the hood and hose (H1, H2), a refeed into a hot, dry coil (H3) and carbon monoxide (H5); the highest design FMEA risk numbers are the feed pump stopping unnoticed (F1), the water seal frozen or dry (F3) and a hood skirt leaking steam in fine soil (F8). Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high; for detection, 5 means hard to detect).

> **Safety:** This register supports the safety sections of STR-PRC-001 and the safety stops of STR-BLD-001; it does not replace them. SteamRoot is a concept on paper; nothing may be built or fired before the written ruling from the boiler authority.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	The authority rules that SteamRoot is a regulated boiler	Texas Health and Safety Code 755.022(a)(4) read strictly: a 0.084 bar seal or a 1.03 bar relief valve not accepted as "atmospheric" or as two independent means	No build as designed; a code boiler would change the pitch	3	5	Project lead (Amish)	Ask for the written ruling first (decided 2026-10-02); present the seal, the relief valve and PR-017; adopt anything stricter it requires
P2	Real beds pass steam too slowly under a light hood	Soil permeability far below sand; skirt leakage	PR-001 and PR-002 missed; burns at the skirt	4	4	Design lead with the co-design partners	Bed preparation note (DEC #18); TRL 4 soil tests choose ballast, skirt depth or steam rate (decided 2026-10-02)
P3	Efficiency below 65 % in use	66.1 % on paper with a 1.1-point margin; dampers left open; baffles left out; firebox exchange or unburned loss worse than modelled (S5, CAL Table 6a)	PR-005 and PR-004 missed; more wood per m2	4	3	Design lead	Baffles in the build plan; damper settings and a flue oxygen reading at TRL 4; PR-004 restatement proposed (DEC #12)
P4	Towed mass over 500 kg	509 kg with both hoods after the baffles and the taller chimney	PR-011 not met	5	2	Design lead	Count PR-011 without the hoods, as towed (DEC #13); mass savings listed in DEC
P5	No co-design partner found	Partners not yet named	Targets not reviewed by users; no site for TRL 4	2	3	Project lead	Texas A&M AgriLife Extension route (decided 2026-10-02)
P6	Cost above what small farms pay	USD 2,720 against the USD 2,200 target	Fewer builds	3	2	Project lead	Savings list in DEC; overrun to USD 2,600 accepted 2026-10-03, USD 120 of STEM fixes reported
P7	Concept choice reopens the design	A and B within 10 % (STR-OPT-001)	Rework if B is chosen	2	3	Project lead (Amish)	Decision #14; B's best idea (feed-loss detection) offered to A either way

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Steam escaping at the hood skirt	Soil resists steam flow more than the hood's weight holds (STEM S11 fails)	Deep burns to feet and hands	4	4	Keep clear of the skirt; boots and gloves; bed preparation; hold-down, skirt or flux from TRL 4 tests (DEC #18)	Medium
H2	Steam at the hose coupling or a lifted hood	Hose moved or hood lifted with steam on	Deep burns	4	3	Divert to the vent first (PR-010); steam-rated hose with whip checks; diverter with no closed position	Medium
H3	Refeed into a hot, dry coil	Pump stops or tank runs dry with the fire burning; feed restarted	Steam and scalding water thrown from joints and the vent at 4.7 times the design flow or more (flash basis open, DEC #20)	5	2	Coil outlet and tank level alarms; latching pump trip with a manual reset (DEC S4); never refeed a hot dry coil; relief valve sized for the flash (PR-009)	Medium
H4	Pressure build-up	Vent blocked, seal pot frozen or overfilled, a valve fitted between coil and vent	Rupture of a joint or the hose	5	1	Open vent that cannot be isolated (PR-006); overflow at the water mark; drain and frost check (decided 2026-10-02); certified relief valve	Low
H5	Carbon monoxide	Incomplete wood combustion, more so at low excess air	Poisoning, death in enclosed spaces	5	2	Firebox and chimney outdoors only; personal CO alarm; tend the fire upwind	Low
H6	Fire and sparks	Embers from the chimney or door; dry vegetation; burn bans	Wildfire, property damage	4	2	6 mm spark arrestor; bare ground or gravel; extinguisher; check burn bans (REG R5)	Low
H7	Hot surfaces	Firebox skin about 100 °C, about 135 °C near the fire bed (STEM S8), flue and chimney hotter, header and pot	Contact burns	3	3	Guarding by distance; gloves; hood handles stay cool (23 °C skin); skin limit and door-side guard proposed (DEC #21)	Medium
H8	Manual handling of the hoods	31.3 kg hood, handles 1.12 m apart (STEM B4 fails)	Back and shoulder injury	3	4	Wheel kit or lifting aid recommended, two-person lift as an interim (DEC S2); ballast lifted off first	Medium until decided

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H9	Soil biology harm	Soil behind the front at 100 °C	Loss of beneficial organisms; manganese and ammonium release	2	4	Do not over-steam; hold time 20 to 30 min; aerated steam as a future option (REVIEW)	Medium
H10	Towing instability or overload	Towing with water aboard (643 kg) or hoods aboard	Loss of control	4	2	Tow drained, hoods on a second vehicle (decided 2026-10-02); axle rated for the full mass	Low
H11	Fibre dust	Cutting and handling ceramic fibre lining	Respiratory irritation	2	3	Mask and gloves in the build plan safety stops	Low

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Feed pump (BOM 4)	Meter 0.5 L/ min into the economizer	Stops or loses prime	Coil runs dry (H3)	Battery flat (E3), air leak, blocked strainer	5	3	3	45	Coil outlet and tank alarms with a latching pump trip (DEC S4); add a feed flow switch (B's best idea, REVIEW); size the battery (DEC #16)
F2	Monotube coil (BOM 6)	Boil 30 kg/h with 5 L of water	Scale blocks the bore or a weld leaks	Higher inlet pressure, hot spots, steam leak	Hard water; dry-out at the top of the helix	4	2	3	24	Rain or softened water; descale; pressure test before use (BLD safety stops)
F3	Water-seal pot (BOM 9)	Limit pressure to 0.084 bar	Frozen, overfilled or dry	Raised relief pressure, or no seal (steam out the vent)	Frost; not refilled after a blow	5	2	4	40	Overflow at the mark; drain daily and in frost; pre-start check (decided 2026-10-02)
F4	Relief valve (BOM 10)	Second line of protection	Stuck shut	No second protection if the seal fails	Corrosion, scale, never tested	5	1	4	20	Lever test as the maker directs; never plug or adjust
F5	Gas-pass baffles (BOM 7, 22)	Make the gas cross the tubes	Missing, warped, misplaced or corroded	Efficiency falls to about 60 % (PR-005)	Left out at assembly; warping in the hot gas; acid condensate in the economizer	3	2	3	18	Build plan steps and sketches 127 and 128; 304 in the bank, 316L in the economizer; inspect when the lid is off
F6	Air dampers (BOM 5)	Hold λ near 1.5	Left open or jammed	Efficiency about 65 % at the λ = 1.6 the draft allows; more wood	Operator habit; distortion	3	3	3	27	Marked damper settings; flue oxygen check at TRL 4
F7	Diverter (BOM 9)	Send steam to hose or vent, never neither	A closed middle position	Coil isolated from the vent (H4)	Wrong valve bought	5	1	2	10	Confirm no closed position when bought (DEC to-confirm 3)
F8	Hood and skirt (BOM 12)	Hold steam on the soil	Steam escapes under the skirt	Burns (H1); slow front; PR-001 missed	Fine soil, uneven bed, hood too light	4	4	2	32	Bed preparation (DEC #18); ballast and skirt depth from TRL 4 tests
F9	Steam hose (BOM 11)	Carry steam 6 m to the hood	Burst or coupling leak	Burns (H2)	Hot-water hose bought instead	5	1	2	10	Steam-rated hose with whip checks;

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
					of steam hose; wear					inspect before each use
F10	Fibre lining (BOM 5)	Keep the shell near 100 °C	Cracked or fallen blanket	Hot spots on the shell; burns	Loading damage, vibration in towing	3	3	3	27	Pins and washers; inspect before lighting
F11	Battery (BOM 13)	Run the pump and alarms	Runs flat in a session	Pump stops (F1); alarms dead	Undersized for the pump chosen (E3)	4	3	3	36	Size from the pump datasheet (DEC #16); charge between fills